STAT201 · STATISTICS · LECTURE 15

Final Review and Exam Preparation

Course summary, a master table of methods, a formula sheet and a practice exam paper

Topic 15 of the syllabus · week 15 · 4 academic hours

LECTURE GOALS

- link the topics of the course into one chain: data → probability → distributions → sampling → estimation → hypothesis testing → regression;
- quickly choose a method and formula from the statement of a problem;
- use the z, t, χ^2, F tables (appendix “Statistical Tables”);
- solve a practice exam in the format of the final assessment and evaluate your own preparation.

1 The running logic of the course

The whole course answers one question: *how to make well-founded decisions under uncertainty*. The main links:

1. **Describing data** (lecture 1): center, spread, shape, outliers, the relationship between two variables.
2. **Probability** (2–3): events, addition and multiplication rules, conditional probability, Bayes’ theorem.
3. **Random variables** (4–6): expectation and variance, binomial, Poisson, normal and other distributions, covariance and correlation.
4. **Linking sample and population** (7–8): sampling distribution, standard error, CLT.
5. **Statistical inference** (9–11): CIs and hypothesis tests for a mean, proportion, difference, categories and several groups.
6. **Models** (12–13): regression, inference, causality.
7. **Practice** (14): study design, report, reproducibility, responsible use of AI.

The general template of inference is the same everywhere: $\dfrac{\text{estimate} - \text{value under } H_0}{\text{SE}}$ is compared with the normal or t distribution. The ability to find the right SE and the right number of degrees of freedom is the core of the course.

2 Master table of standard errors and tests

Parameter	Standard error	Distribution	Degrees of freedom
μ (σ known)	$\sigma/\sqrt{n}$	$N(0,1)$	—
μ (σ unknown)	$s/\sqrt{n}$	t	$n-1$
p	$\sqrt{p_0(1-p_0)/n}$ (test); $\sqrt{\hat{p}(1-\hat{p})/n}$ (CI)	$N(0,1)$	—
$\mu_1-\mu_2$ (indep.)	$\sqrt{s_1^2/n_1+s_2^2/n_2}$	t (Welch)	Welch formula
μ_d (pairs)	$s_d/\sqrt{n}$	t	$n-1$
p_1-p_2	$\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1}+\frac{1}{n_2})}$ (test)	$N(0,1)$	—
β_1	$s_e/\sqrt{S_{xx}}$	t	$n-2$
$r\times c$ table	$\sum(O-E)^2/E$	χ^2	$(r-1)(c-1)$
k means	$F=MSB/MSW$	F	$k-1;\ N-k$

3 Formula sheet of the course

Data description and probability

$$\bar{x}=\tfrac{1}{n}\sum x_i,\quad s^2=\tfrac{1}{n-1}\sum(x_i-\bar{x})^2,\quad CV=\tfrac{s}{\bar{x}},\quad IQR=Q_3-Q_1;$$

outlier fences: $Q_1-1.5\,IQR$ and $Q_3+1.5\,IQR$;

$$P(A\cup B)=P(A)+P(B)-P(A\cap B);\quad P(A\cap B)=P(A)P(B\mid A);$$

$$P(A\mid B)=\tfrac{P(A\cap B)}{P(B)};\quad C_n^k=\tfrac{n!}{k!(n-k)!};\quad P(A_i\mid B)=\tfrac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)};$$

odds: $\text{odds}_{\text{post}}=\text{odds}_{\text{prior}}\cdot LR.$

Random variables and distributions

$$\mu=\sum xp(x),\quad \sigma^2=\sum x^2p(x)-\mu^2;\quad \mathrm{E}[aX+b]=a\mu+b,\quad \mathrm{Var}(aX+b)=a^2\sigma^2;$$

$$\mathrm{Bin}(n,p):\ \mu=np,\ \sigma^2=np(1-p);\qquad \mathrm{Pois}(\lambda):\ \mu=\sigma^2=\lambda;$$

$$\mathrm{Exp}(\lambda):\ \mu=\tfrac{1}{\lambda};\qquad U(a,b):\ \mu=\tfrac{a+b}{2},\ \sigma^2=\tfrac{(b-a)^2}{12};$$

$$z=\tfrac{x-\mu}{\sigma};\quad \mathrm{Cov}(X,Y)=\mathrm{E}[XY]-\mu_X\mu_Y;\quad \rho=\tfrac{\mathrm{Cov}}{\sigma_X\sigma_Y};$$

$$\mathrm{Var}(X\pm Y)=\mathrm{Var}\,X+\mathrm{Var}\,Y\pm 2\,\mathrm{Cov}\,.$$

Inference and regression

$$\mathrm{SE}(\bar X)=\tfrac{\sigma}{\sqrt{n}};\quad \bar x\pm t_{\alpha/2,n-1}\tfrac{s}{\sqrt{n}};\quad \hat p\pm z_{\alpha/2}\sqrt{\tfrac{\hat p(1-\hat p)}{n}};$$

$$n\ge\left(\tfrac{z\sigma}{E}\right)^2;\quad n\ge\tfrac{z^2p(1-p)}{E^2};$$

$$t=\tfrac{\bar x-\mu_0}{s/\sqrt{n}};\quad z=\tfrac{\hat p-p_0}{\sqrt{p_0(1-p_0)/n}};\quad t=\tfrac{\bar x_1-\bar x_2}{\sqrt{s_1^2/n_1+s_2^2/n_2}};\quad t=\tfrac{\bar d}{s_d/\sqrt{n}};$$

$$\chi^2=\sum\tfrac{(O-E)^2}{E};\quad F=\tfrac{SSB/(k-1)}{SSW/(N-k)};$$

$$b_1=\tfrac{S_{xy}}{S_{xx}}=r\tfrac{s_y}{s_x};\quad b_0=\bar y-b_1\bar x;\quad R^2=\tfrac{SSR}{SST};\quad s_e=\sqrt{\tfrac{SSE}{n-2}};\quad \mathrm{SE}(b_1)=\tfrac{s_e}{\sqrt{S_{xx}}};$$

$$\hat y_0\pm t\,s_e\sqrt{1+\tfrac{1}{n}+\tfrac{(x_0-\bar x)^2}{S_{xx}}}.$$

4 Typical exam mistakes

- Mixing up $\sigma/\sqrt{n}$ (mean), $\sigma\sqrt{n}$ (sum) and σ (a single observation).
- Not checking the conditions of applicability (normal approximation, independence, $n\ge 30$, expected counts ≥ 5).
- z instead of t when σ is unknown and n is small; the wrong number of degrees of freedom.
- The wrong direction of the alternative and a wrong p-value (doubling for a two-sided test).
- Interpreting a CI or p-value as the "probability of the hypothesis"; concluding that "H_0 was proved".
- Analyzing paired data as independent, and vice versa.
- Extrapolating a regression; drawing causal conclusions from correlation.
- Missing units and a verbal interpretation in the answer.

5 How to prepare and how to take the exam

1. Solve the practice paper (below) in 120 minutes without hints, then check against the solutions and analyze each mistake: which topic does it belong to.
2. Make your own one-page "formula sheet" (even if tables are allowed); this is the best way to memorize formulas.
3. In the exam: read all the problems; start with those you know how to do; in each one write "given / find / method / computation / conclusion in words".
4. Check reasonableness: probability in $[0,1]$, variance ≥ 0, $|r|\le 1$, a CI symmetric about the estimate, a p-value consistent with the decision and the CI.
5. Round only at the end; keep intermediate values to 4–5 digits.

6 Key terms of the course

Term	*In Russian*	Meaning
Sampling distribution	*Выборочное распределение*	distribution of a statistic
Standard error	*Стандартная ошибка*	SD of a statistic
Confidence interval	*Доверительный интервал*	estimate $\pm$ critical$\times$SE
p**-value**	*p-значение*	$P(\text{at least as extreme} \mid H_0)$
Power	*Мощность*	$1-\beta$
Residual	*Остаток*	$y-\hat{y}$
Confounder	*Искажающая переменная*	related to x and y

7 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Monthly sales (million UZS) of eight managers: $14,\ 18,\ 20,\ 22,\ 25,\ 27,\ 31,\ 63$.

(a) Find the mean, median and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Which measure of center is preferable and why? How does the mean change without the largest value?

Problem 2 [B] The default rate among borrowers is 4%. A scoring model flags 80% of future defaulters and 10% of reliable borrowers as “risky”. A borrower is flagged. What is the probability that he will default? What does the probability become if the borrower is flagged by two independent scoring models with the same characteristics?

Problem 3 [B] A customer accepts an offer with probability 0.25, independently of others; the offer is made to 12 customers.

(a) Find the expected number of acceptances and the standard deviation.
(b) Find the probability that at least 5 accept.
(c) Find the probability that exactly 3 accept.

Problem 4 [A] Test scores $X \sim N(65, 8^2)$. Find $P(X > 75)$, $P(60 < X < 70)$ and the 85th percentile.

Problem 5 [B] Number of purchases per customer: $\mu = 2.5$, $\sigma = 1.2$. A sample of $n = 64$ customers. Find $\mathrm{SE}(\bar{X})$ and $P(\bar{X} > 2.7)$. Explain why the normal distribution can be used.

Problem 6 [B] A sample of 20 customers: $\bar{x} = 48$ min, $s = 9$. (a) Construct a 95% CI for the mean. (b) Test $H_0 : \mu = 45$ against $H_1 : \mu \neq 45$ ($\alpha = 0.05$). (c) Are the results of (a) and (b) consistent?

Problem 7 [B] In a sample of 250 customers, 62% use the app. (a) Construct a 95% CI for the proportion. (b) How many customers must be surveyed for the margin of error to be 4 percentage points (with the proportion unknown)?

Problem 8 [B] Processing time (min) for 7 employees before and after a new system was introduced: before $30, 28, 35, 40, 25, 33, 38$; after $27, 26, 33, 36, 25, 30, 35$. Test whether processing time decreased (two-sided test, $\alpha = 0.05$) and construct a 95% CI for the mean change.

Problem 9 [B] A survey of 350 customers: age group and app use.

	Use the app	Do not use
Under 30	60	40
30–45	50	70
Over 45	40	90

Test independence ($\alpha = 0.05$). Which group contributes most to χ^2?

Problem 10 [B] ANOVA for 4 groups of 5 observations each: $SSB = 84$, $SSW = 160$. Complete the table and draw a conclusion ($\alpha = 0.05$).

Problem 11 [B] For $n = 10$ observations: $\bar{x} = 20$, $\bar{y} = 50$, $S_{xx} = 400$, $S_{xy} = 600$, $S_{yy} = 1200$.

(a) Find the regression equation, R^2 and s_e.
(b) Test the significance of the slope and construct a 95% CI for β_1.
(c) Predict y at $x = 25$ and construct a 95% prediction interval.

Problem 12 [C] A researcher found a positive significant relationship between the number of training sessions held by a bank and the mean income of customers in the region ($r = 0.62$, $p < 0.01$) and proposed increasing the number of training sessions to raise income. Give a critical assessment of the conclusion and suggest how causality could be tested.

Problem 13 [C] A manager wants to know whether customer satisfaction (rating 1–5) differs across three branches and chooses to "run three pairwise t-tests A–B, A–C, B–C at the 0.05 level". Why is this wrong? What should be proposed? What value of α should be used for each pairwise comparison under the Bonferroni correction?

Problem 14 [C] Carry out a mini-analysis. In data on 200 customers the mean spending is $\bar{x} = 4.8$ million, $s = 1.5$. (a) Construct a 95% CI for the mean. (b) The bank claims $\mu = 5$. Test it. (c) What is the effect size $d = (\bar{x} - \mu_0)/s$? (d) Is $n = 200$ sufficient to detect a deviation of 0.2 million with 80% power?